

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ ls /etc/ > file.txt  
gakuligin@gakuligin:~$ ls >> file.txt  
gakuligin@gakuligin:~$ cat file.txt  
abrt  
adjtime  
aliases  
alsa  
alternatives  
anaconda  
anacrontab  
anthy-unicode.conf  
asound.conf  
at.deny  
audit  
authselect  
avahi  
bash_completion.d  
bashrc  
bindresvport.blacklist  
binfmt.d  
bluetooth  
brlapi.key  
brltty  
brltty.conf  
ceph  
chromium  
chrony.conf  
cifs-utils  
cockpit  
colord  
containers
```

Рисунок 1: Запись в файл

```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ grep .conf file.txt > conf.txt  
gakuligin@gakuligin:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprind.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf  
locale.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage03/report/_resources/csl
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage03/report/bib/cite.bib
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage04/report/_resources/csl
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage04/report/bib/cite.bib
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage05/report/_resources/csl
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage05/report/bib/cite.bib
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage06/report/_resources/csl
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/project-personal/stage06/report/bib/cite.bib
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/presentation/report/_resources/csl
/home/gakuligin/work/study/2025-2026/Операционные системы/2026-1--study--os-in
tro/presentation/report/bib/cite.bib
/home/gakuligin/git-extended/.git/hooks/commit-msg.sample
/home/gakuligin/git-extended/.git/objects/97/c06c5576c3415c7535ff7d173fa8f6ce7
7af66
/home/gakuligin/git-extended/.git/objects/11/c0cb8993753a95d54ca1dbd3415ee18ff
93f36
/home/gakuligin/git-extended/.git/config
/home/gakuligin/conf.txt
gakuligin@gakuligin:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashes.d/hashes.new
find: /etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
'/etc/libvirt': Permission denied
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/ssh_config.d': Permission denied
find: '/etc/sss': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
/etc/systemd/system/httpd.service.d
/etc/systemd/system/multi-user.target.wants/happd.service
/etc/systemd/system/happd.service
/etc/udev/hwdb.d
/etc/udev/hwdb.bin
:
```

Рисунок 4: Поиск файлов

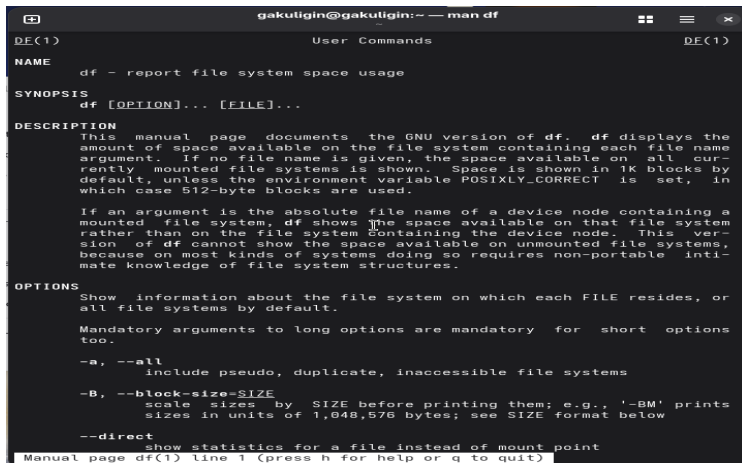
```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ find ~ -name "log*" > logfile &  
[1] 6400  
gakuligin@gakuligin:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
gakuligin@gakuligin:~$ rm logfile  
gakuligin@gakuligin:~$
```

Рисунок 5: Фоновый запуск процесса



```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ gedit &  
[1] 6435  
gakuligin@gakuligin:~$ ps | grep gedit  
    6435 pts/0    00:00:00 gedit  
gakuligin@gakuligin:~$ kill 6435  
[1]+  Terminated                  gedit  
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$
```

Рисунок 6: Фоновый запуск и завершение процесса



```
gakuligin@gakuligin:~ — man df
DE(1)                                User Commands                                DE(1)

NAME
df - report file system space usage

SYNOPSIS
df [OPTION]... [FILE]...

DESCRIPTION
This manual page documents the GNU version of df.  df displays the
amount of space available on the file system containing each file name
argument.  If no file name is given, the space available on all cur-
rently mounted file systems is shown.  Space is shown in 1K blocks by
default, unless the environment variable POSIXLY_CORRECT is set, in
which case 512-byte blocks are used.

If an argument is the absolute file name of a device node containing a
mounted file system, df shows the space available on that file system
rather than on the file system containing the device node.  This ver-
sion of df cannot show the space available on unmounted file systems,
because on most kinds of systems doing so requires non-portable inti-
mate knowledge of file system structures.

OPTIONS
Show information about the file system on which each FILE resides, or
all file systems by default.

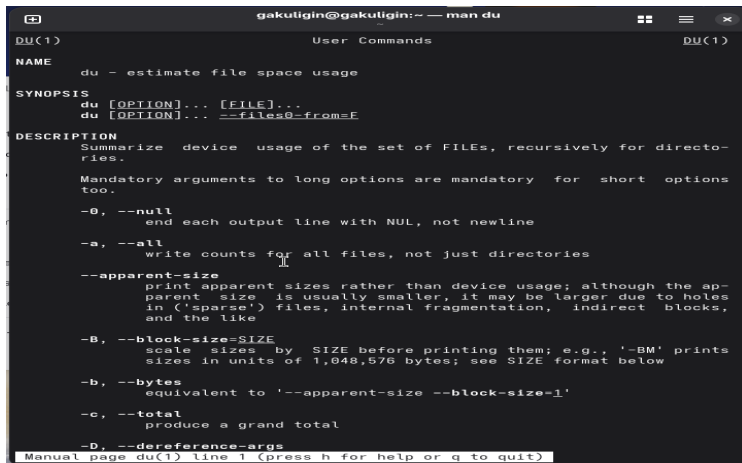
Mandatory arguments to long options are mandatory for short options
too.

-a, --all
    include pseudo, duplicate, inaccessible file systems

-B, --block-size=SIZE
    scale sizes by SIZE before printing them; e.g., '-BM' prints
    sizes in units of 1,048,576 bytes; see SIZE format below

--direct
    show statistics for a file instead of mount point
Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df



```
gakuligin@gakuligin:~ — man du
DU(1)                                User Commands                                DU(1)
NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    -b, --bytes
        equivalent to '--apparent-size --block-size=1'

    -c, --total
        produce a grand total

    -D, --dereference-args
        Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ df  
Filesystem      1K-blocks      Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 89965540  64564716  59% /  
devtmpfs         4011280          0    4011280   0% /dev  
tmpfs            4035552          92    4035560   1% /dev/shm  
tmpfs            1614254        1952    1612312   1% /run  
tmpfs            1024             0       1024    0% /run/credentials/systemd-jour  
nald.service  
tmpfs            4035656        1612    4034044   1% /tmp  
/dev/nvme0n1p3 155186176 89965540  64564716  59% /home  
/dev/nvme0n1p2 1992552    518800    1352512  28% /boot  
/dev/loop2      49280       49280          0 100% /var/lib/napd/snap/snapd/259  
35  
/dev/loop1      115712     115712          0 100% /var/lib/napd/snap/hugo/2567  
6  
/dev/loop0       68480      68480          0 100% /var/lib/napd/snap/core24/13  
49  
tmpfs           1024          0       1024    0% /run/credentials/systemd-reso  
lved.service  
tmpfs           807128        168    806960   1% /run/user/1038  
/dev/sr0        2677920     2677920          0 100% /run/media/gakuligin/Fedora-W  
S-Live-43  
gakuligin@gakuligin:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/5c
4      ./git-extended/.git/objects/00
4      ./git-extended/.git/objects/b9
4      ./git-extended/.git/objects/97
4      ./git-extended/.git/objects/38
4      ./git-extended/.git/objects/11
4      ./git-extended/.git/objects/1a
4      ./git-extended/.git/objects/58
4      ./git-extended/.git/objects/f9
4      ./git-extended/.git/objects/0a
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
834488 .
gakuligin@gakuligin:~$
```

Рисунок 10: Запуск команды du


```
/home/gakuligin/git-extended/.git/objects/67
/home/gakuligin/git-extended/.git/objects/af
/home/gakuligin/git-extended/.git/objects/5c
/home/gakuligin/git-extended/.git/objects/00
/home/gakuligin/git-extended/.git/objects/b9
/home/gakuligin/git-extended/.git/objects/97
/home/gakuligin/git-extended/.git/objects/38
/home/gakuligin/git-extended/.git/objects/11
/home/gakuligin/git-extended/.git/objects/1a
/home/gakuligin/git-extended/.git/objects/58
/home/gakuligin/git-extended/.git/objects/f9
/home/gakuligin/git-extended/.git/objects/0a
/home/gakuligin/git-extended/.git/logs
/home/gakuligin/git-extended/.git/logs/refs
/home/gakuligin/git-extended/.git/logs/refs/heads
/home/gakuligin/git-extended/.git/logs/refs/remotes
/home/gakuligin/git-extended/.git/logs/refs/remotes/origin
/home/gakuligin/monthly
/home/gakuligin/reports
/home/gakuligin/reports/monthly
/home/gakuligin/reports/monthly/monthly
/home/gakuligin/ski.places
/home/gakuligin/ski.places/equipment
/home/gakuligin/ski.places/plans
/home/gakuligin/australia
/home/gakuligin/play
/home/gakuligin/play/games
/home/gakuligin/play/games/play
gakuligin@gakuligin:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.